9.2.3

$k = N_1/N_2 = 2080/80 = 26.$

$U_{1P} = U_{1L}/\sqrt{3} = 6000/\sqrt{3}\ V \approx 3464.1V$

$U_{2P} = U_{1P}/k = 3464.1/26 \approx 133.2V$

$U_{2L} = \sqrt{3}U_{2P} = \sqrt{3}\times 133.2V \approx 230.7V.$

$U_{2L} = U_{2P} = 133.2V$

9.2.4

(1) $N_{21} = \frac{U_{21}}{U_1}N_1 = 230$

$N_{22} = \frac{U_{22}}{U_1}N_1 = 75$

$N_{23} = \frac{U_{23}}{U_1}N_1 = 13$

(2) $S = U_{21}I_{21} + U_{22}I_{22} + U_{23}I_{23}$

$= 110\times 0.2 + 36\times 0.5 + 6.3\times 1 = 46.3V\cdot A.$

$I_1 = S/U_1 = 46.3/220 = 0.21A.$

9.2.5

$R_L' = k^2R_L = (\frac{N_1}{N_2})^2R_L = (\frac{300}{100})^2\times 8\Omega = 72\Omega$

$P = (\frac{U_S}{R_S+R_L'})^2R_L' \approx 0.088W.$

9.3.3

(1) $I_N = P_N/(\sqrt{3}U_N\eta_N\cos\varphi_N) = 84.2A.$

(2) $S_N = (n_1-n_N)/n_1 = 0.013.$

(3) $T_N = 9550\,P_N/n_N = 290.4\,N.m.$

$T_{max} = 2.2T_N = 638.9\,N.m.$

$T_S = 1.9T_N = 551.8\,N.m.$

9.3.4

(1) $I_S = 7I_N = 107.8A$

(2) $T_S' = \frac{1}{3}T_S = \frac{22}{30}T_N = \frac{22}{30}\times 9550\cdot\frac{P_N}{n} = 49.7N\cdot m.$

$\therefore T_S' = 36.4N\cdot m.$

9.3.5

(1) $\eta = \frac{P_2}{P_1} \times 100\% = 83.3\%$.

$\cos\varphi_1 = P_1 / (\sqrt{3} U_1 I_1) \approx 0.82$

(2) $\eta = P_2 / P_1 \times 100\% \approx 62.5\%$.

$\cos\varphi_1 = P_1 / (\sqrt{3} U_1 I_1) = 0.51$.

9.3.7

$T_N = 9550 \cdot \frac{P_N}{n_N} = 108.3 \ N \cdot m$.

$\therefore T_{max} = 2 T_N = 216.6 \ N \cdot m$